ESP32 Water Meter Reader

Handover and instructions · version 2 · 15 September 2026

1. What it does

- **Live meter reading:** an ESP32-CAM photographs the water meter and sends the photo over Wi-Fi to a computer. The computer reads the counter and saves the reading.
- **Whole-dataset reading:** give it a ZIP, a folder, or many photos and it reads every one.
- **Excel and CSV export:** every reading can be downloaded as an Excel workbook (Readings + Summary sheets) or a CSV file.
- **Checking against known values:** if the dataset includes the correct readings (for example Kaggle's `data.csv`), every photo is marked *Match: Yes / No* and accuracy is calculated.
- **Runs offline** on an ordinary Windows or Linux computer. No internet, cloud account or graphics card is needed after installation.

2. Results on the Kaggle Water Meters dataset

The reader uses two small models trained on the 1,244-photo Kaggle dataset: one finds the counter, the other reads its 8 digits. 196 photos were kept out of training and used only for testing, so these figures show what to expect on new photos:

77%
all 8 digits exactly right (151 / 196)

90%
whole cubic metres right

99.5%
counter found in the photo

0.1 s
per photo on a laptop CPU

For comparison, the previous version (general-purpose OCR) read about **3%** of these meters correctly and took about 7 seconds per photo.

How to use the confidence score

Confidence	Share of photos	Readings correct	What to do
75% or more (green)	52%	**97%**	Trust it
45–75% (amber)	22%	**88%**	Usually right; spot-check
Below 45% (red)	26%	**27%**	Check by eye (use the *Low confidence* filter)

All 1,244 dataset photos were also processed through the dashboard with 0 failures and 1,043 exact matches (83.8%). That overall figure is higher than 77% because most photos were used in training. The exports label each photo as *train*, *val* or *test* for this reason.

3. What is in the package

Item	Purpose
`install_windows.bat`, `run_windows.bat`	One-time install and start on Windows
`run_linux.sh`	Install and start on Linux
`app/`	The dashboard and reading engine
`models/`	The two trained models (4.6 MB) and their accuracy report
`batch_read.py`	Command-line tool: dataset folder/ZIP → Excel + CSV
`results/`	Example exports already produced from the Kaggle dataset (full dataset + test photos)
`esp32/esp32_cam_meter_sender.ino`	Arduino sketch for the ESP32-CAM
`training/`, `requirements-train.txt`	Retrain the models on new photos (optional)
`tests/`	Automated tests (15)
`README.md`	Full technical documentation

4. Install and start

Windows

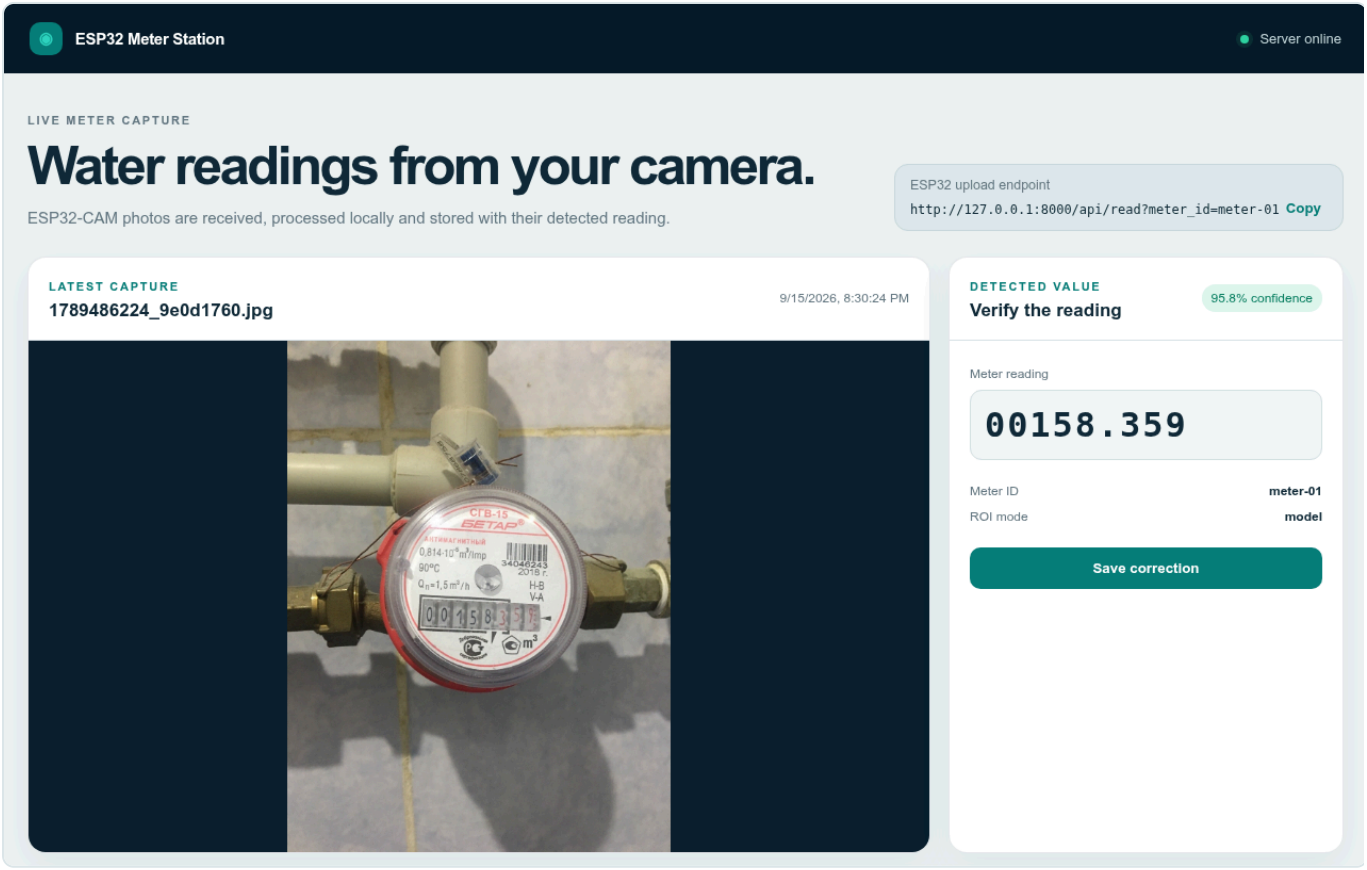
1. Install 64-bit **Python 3.11 or 3.12** from python.org and tick **Add Python to PATH**.
2. Extract the ZIP.
3. Double-click `install_windows.bat` once (needs internet, a few minutes).
4. Double-click `run_windows.bat` whenever you want to use it. The dashboard opens in the browser.
5. If Windows asks about the firewall, allow **Private networks** so the ESP32 can connect.

Linux

```
cd WaterMeterESP32Software
./run_linux.sh
```

5. Where to see it

Open **`http://127.0.0.1:8000`** in a browser on the same computer. Other devices on the same Wi-Fi use the computer's IP address instead, e.g. `http://192.168.1.20:8000`.

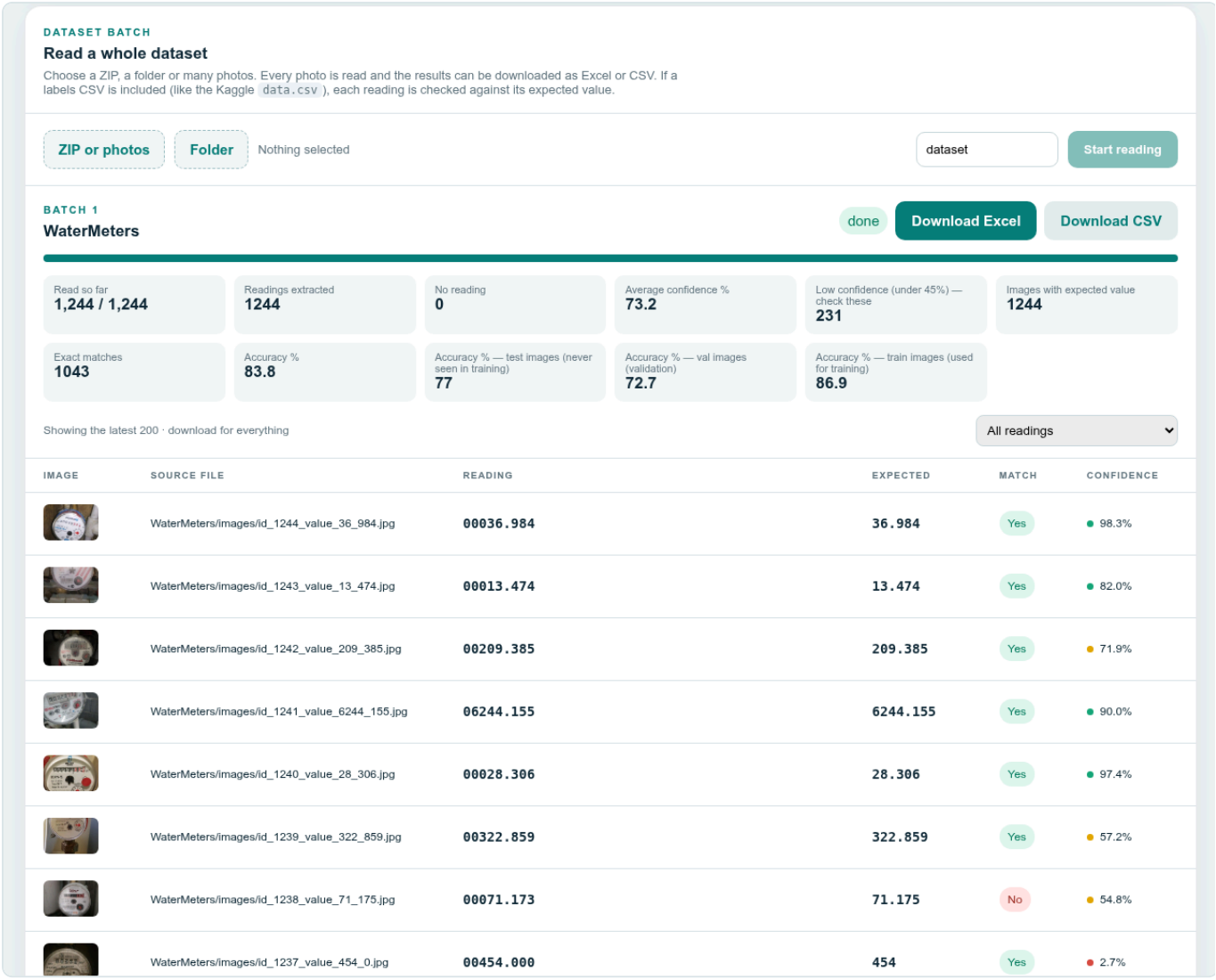


Live view: the latest photo and its detected reading (here 00158.359 at 95.8% confidence). A wrong reading can be corrected and saved.

6. Read a dataset and export to Excel / CSV

From the dashboard

1. Scroll to **Read a whole dataset**.
2. Click **ZIP or photos** or **Folder** and choose the dataset. For Kaggle, choose the `WaterMeters` folder; `masks` and `collage` are skipped automatically.
3. Press **Start reading**. Progress, accuracy and readings update live.
4. Press **Download Excel** or **Download CSV**. Use the filter to show *Low confidence* or *Mismatches*.



The full Kaggle dataset: 1,244 photos read, with a summary and every reading compared with its expected value.

From the command line (no browser needed)

```
Windows:  .venv\Scripts\python batch_read.py C:\path\to\WaterMeters
Linux:    .venv/bin/python batch_read.py ~/Downloads/water-meters-dataset/WaterMeters
```

This writes `results/WaterMeters.xlsx` and `results/WaterMeters.csv`. Useful options: `--limit 50` (quick trial), `--subset test` (only photos the model never trained on), `--out folder`, `--name run1`.

Excel columns

Source file · *Reading* (text, keeps leading zeros) · *Value* (number) · *Confidence %* · *Expected* · *Match* · *Difference* · *Model split* · *Note* (for unreadable files) · *Captured at*. The **Summary** sheet has totals, accuracy and the low-confidence count.

The Kaggle dataset

Download from Kaggle: `kaggle.com/datasets/tapakah68/yandextoloka-water-meters-dataset`, about 1 GB zipped. It contains `data.csv` (file name + correct reading), `images/` and `masks/`. The dataset is not included in this package because of its size.

7. Connect the ESP32-CAM

1. Open `esp32/esp32_cam_meter_sender.ino` in Arduino IDE and select board **AI Thinker ESP32-CAM**.
2. Enter the Wi-Fi name and password. In `SERVER_URL`, replace `192.168.1.100` with the computer's IPv4 address (from `ipconfig`).
3. Upload with GPIO 0 connected to GND, then remove the jumper and reset. By default the camera sends a photo every 30 seconds.

Mount the camera straight on to the meter with even light. The meter's 3 red digits are read as decimals (00123.456 m³); this can be changed under *Fixed-camera calibration → Decimal places*.

8. Limits and notes

> The 77% accuracy was measured on Kaggle photos of common Russian domestic meters. Photos from a specific ESP32-CAM installation, other meter models or poor lighting may score lower. Test on your own photos first. For billing-grade use, collect a few hundred labelled photos of the actual meters and retrain (README, "Retraining").

- About 4 in 5 wrong readings on the test photos had confidence below 45%, so reviewing the red rows catches most errors. About a third of wrong readings were within 10 litres of the true value.
- Data stays on the computer, in `data/` (database and photos). Old dataset batches can be deleted from the dashboard.